The finite local amalgamation theory of strong-EQ RCC5 networks

2026-07-16

Abstract

We collect a self-contained, exhaustively machine-checked *local* theory of strong-EQ atomic RCC5 networks: a normal form identifying such networks with ordered-disjoint structures (a strict partial order plus a downward-closed disjointness relation), free amalgamation in that normal form, an exact one-point extension criterion, and exact characterizations of uniform and arbitrary non-uniform cross-pocket guard policies. The normal form's forward direction is machine-*certified* in Lean 4 (`formal/RCC5NormalForm.lean`, axioms: `propext` only); the remaining results are exhaustively verified by finite computation. We are deliberately clear about the boundary of this theory: it is complete *local* algebra and it does *not* yield decidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$. The residual keystone (a global bounded-cover / bounded-width property) is provably not a local composition problem, and remains open. Provenance: this theory was distilled from a cold regular-cover attack (GPT-5.5) on the $\mathcal{ALCI}_{\mathrm{RCC5}}$ decidability problem; we record it with its verification status attached.

1 Setting

The RCC5 relation alphabet is $\{\mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}\}$. Under strong-EQ semantics EQ is identity, so between distinct elements exactly one of $\mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}$ holds; converse is $\mathsf{PP}^{-1} = \mathsf{PPI}$, $\mathsf{PPI}^{-1} = \mathsf{PP}$, $\mathsf{PO}^{-1} = \mathsf{PO}$, $\mathsf{DR}^{-1} = \mathsf{DR}$. A network on a set is a total labelling ρ of ordered pairs by atoms. It is a (strong-EQ) RCC5 network if it is reflexively EQ, has EQ only on the diagonal, is converse-closed, and is composition-closed: $\rho(x, z) \in \mathrm{comp}(\rho(x, y), \rho(y, z))$ for all x, y, z, over the RCC5 composition table.

2 The normal form

Definition 1 (Ordered-disjoint structure)**.** A set with a strict partial order $<$ and a symmetric irreflexive relation $\#$ is *ordered-disjoint* if (i) $x < y \Rightarrow \neg\, x\#y$, and (ii) $\#$ is downward closed: $x\#y$, $x' \leq x$, $y' \leq y \Rightarrow x'\#y'$ (where $\leq$ is $<$ or $=$).

Theorem 2 (RCC5 normal form)**.** *For strong-EQ atomic networks, RCC5 composition closure is equivalent to an ordered-disjoint structure via*

$$x < y \iff x \,\mathsf{PP}\, y, \qquad x\#y \iff x \,\mathsf{DR}\, y,$$

with PPI *the converse of* PP *and* PO *residual: for distinct* x, y, $x \,\mathsf{PO}\, y$ *iff* $\neg(x < y)$, $\neg(y < x)$, $\neg(x\#y)$.

The forward direction — *every RCC5 network is ordered-disjoint* — rests on exactly three composition cells:

$$\mathrm{comp}(\mathsf{PP},\mathsf{PP}) = \{\mathsf{PP}\} \text{ (transitivity)}, \quad \mathrm{comp}(\mathsf{PP},\mathsf{DR}) = \{\mathsf{DR}\}, \quad \mathrm{comp}(\mathsf{DR},\mathsf{PPI}) = \{\mathsf{DR}\} \text{ (downward closure)},$$

together with converse and strong-EQ reflexivity. It is machine-*certified* in Lean 4: `formal/RCC5NormalForm.lean` proves, for an arbitrary domain and any `RCC5Net`, that the PP/DR read-off satisfies all six ordered-disjoint axioms (`toOrderedDisjoint`), with the strict-order and downward-closure facts exposed as corollaries (`pp_strict_order`, `dr_downward_closed`); the file depends on `propext` only and has zero `sorry`. The converse direction (an ordered-disjoint structure's read-off is composition-closed) is exhaustively checked by finite computation (`wp47`: all complete atomic networks up to $n = 4$, 916/916 composition-closed networks matching the characterization, 0 mismatches) and is a natural formalization follow-up.

3 Amalgamation and extension

Theorem 3 (Free amalgamation)**.** *Ordered-disjoint structures freely amalgamate over a common induced substructure: taking* $<^*$ *the transitive closure of the union of the orders and* $\#^*$ *the downward closure of the union of the disjointness relations along* $<^*$ *yields an ordered-disjoint structure whose restriction to each factor is the original.*

(Verified: `wp48`.) The proof compresses any putative order cycle or comparable-disjoint conflict into one factor, where it is impossible.

Proposition 4 (One-point extension)**.** *Let* P *be a finite closed pocket and add one connector* e *with* $U = \{x : e\,\mathsf{PP}\,x\}$, $L = \{x : e\,\mathsf{PPI}\,x\}$, $D = \{x : e\,\mathsf{DR}\,x\}$. *The extension is RCC5-consistent iff* U *is upward closed,* L *downward closed,* D *downward closed,* $x \in U \wedge x\#y \Rightarrow y \in D$, *and every* $\ell \in L$ *lies below every* $u \in U$ *and is disjoint from every* $d \in D$ *already in* P.

(Verified against brute-force closure: `wp71`, pockets up to size 4.) Finite multi-connector topology reduces to sequential one-point traces, and every finite pocket admits a canonical principal connector basis of size $2|P| + 1$.

Lemma 5 (Ghost-connector elimination)**.** *Guard-only connectors — connectors that are neither existential witnesses nor otherwise formula-visible — may be erased: an induced subnetwork of a closed network is closed, so such connectors compile into direct pair policies among the remaining nodes.*

(Verified: `wp81`.)

4 Cross-pocket policies

Theorem 6 (Uniform cross-policy)**.** *For finite closed pockets* A, B*: all-cross* DR *and all-cross* PO *are always RCC5-safe; all-cross* $A < B$ *(every* $a\,\mathsf{PP}\,b$*) is safe iff* B *has no internal* DR *pair; all-cross* $B < A$ *is safe iff* A *has no internal* DR *pair.*

(Verified: `wp82`, exhaustive to 3×3, 0 mismatches.)

Theorem 7 (Non-uniform cross-policy)**.** *For finite closed pockets* A, B *and an arbitrary atom assignment* χ *to all cross pairs,* $A \cup B \cup \chi$ *is a strong-EQ composition-closed amalgam preserving* *both pockets iff the six ordered-disjoint closure conditions hold:* $<^*$ *(transitive closure of the orders plus cross* PP/PPI*) is irreflexive and restricts to each pocket's order;* $\#^*$ *(downward closure of the disjointness plus cross* DR*) is irreflexive and restricts to each pocket's disjointness; no comparable pair is disjoint; and* χ *agrees with the atoms read off* $<^*, \#^*$.

(Verified against brute force: `wp83`, 0 mismatches on exhaustive small cases and 2500 random larger tests.)

Together, Theorems 2–7 with Proposition 4 and Lemma 5 *exhaust the local algebra*: arbitrary finite guard policies among finite pockets are checkable by finite order/disjointness closure, and no higher RCC5-specific arity than the ternary composition is needed (RCC5 composition is a finite set of forbidden edge-coloured triangles).

5 The boundary of this theory

We state plainly what this local theory does *not* give.

Remark 8 (No decidability)**.** Theorem 2 and its companions characterize *when a given network is RCC5-consistent* and *how finite pieces amalgamate*. They do not bound the size of a finite presentation realizing a given $\mathcal{ALCI}_{\mathrm{RCC5}}$ concept. Concept satisfiability additionally requires a global property — that every satisfiable concept has a bounded coherent regular cover / bounded live-width presentation — which is the standing open keystone (equivalently the bounded-width property F6). The local theory is on the settled side of the local/global divide.

Remark 9 (Why standard covers do not close it)**.** Because RCC5 relations are total, the Gaifman graph of every nontrivial model is complete, and even a one-dimensional PP-chain has finite prefixes of unbounded clique number and treewidth (`wp78`). Hence off-the-shelf guarded / clique-guarded bounded-treewidth cover theorems do not apply; the keystone would require a bespoke cover theorem for complete edge-coloured regular structures. This is exactly the two-decade obstruction, unmoved by the local theory above.

6 Provenance and status

This theory was distilled from a cold regular-cover attack (GPT-5.5) on $\mathcal{ALCI}_{\mathrm{RCC5}}$ decidability (packages under `papers/regular_cover_route_pivot`; probes `wp45`–`wp83`). The normal form's forward direction is Lean-certified (`RCC5NormalForm.lean`); every other result here is exhaustively machine-checked by finite computation but is *theorem-level, not kernel-certified*, and should be independently reviewed (and ideally Lean-transcribed) before being cited as established. The one $\mathcal{ALCI}_{\mathrm{RCC5}}$-level corollary in reach — decidability of the ∀PO-free fragment — is a re-derivation of an already known result (the two-tier PO-coherent fragment; see `papers/po_free_fragment_ALCIRCC5.tex` and `papers/two_tier_quotient_ALCIRCC5.tex`, Remark on automatic PO-coherence). Consistent with the project's standing discipline, the honest label is *strongly supported, not certified*, with the normal form's forward direction the exception (certified).